

Sound Data-Flow Verification for Safe Serverless Workflow Orchestration: Provenance, Typestate, Concurrency, and Compensation

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Abstract. Serverless workflow orchestration platforms such as AWS Step Functions coordinate fleets of functions and managed services, but their workflows are written as untyped JSON state machines (the Amazon States Language, ASL) whose correctness is checked only at run time. The validators developers have today—`statelint`, `asl-validator`, and AWS’s own `ValidateStateMachineDefinition`—check JSONPath *syntax*, not whether the field a path names can ever have been *produced*, whether tasks run in a valid business order, whether a retried operation is safe to repeat, whether concurrent branches contend on a resource, or whether a retry can complete within its timeout. We present STEP CHECK, a static verifier whose centrepiece is, to our knowledge, the first *sound*, flow-sensitive *shape*-level data-flow (field-provenance) analysis for a deployed serverless workflow language: an abstract interpretation of the ASL I/O-processing pipeline over a may-present lattice of document shapes that reports a missing-field reference only when it is guaranteed to fail at run time. We prove this no-false-positive property as a soundness theorem; it recovers the classic “`amount` vs. `total`” contract bug with zero annotations. Where the data-flow-anomaly checkers studied for abstract business-process models reason over coarse create/read element abstractions of hand-built nets, our analysis tracks concrete nested JSON document shapes and runs on—and recompiles to—the real deployed artifact; cast in service terms, its typed tier is a sound static *consumer-driven-contract* check across orchestrated services. Around it STEP CHECK adds seven further checks—structural well-formedness, typed contracts, typestate, retry-safety, *concurrency-interference*, *retry/timeout-budget*, and Saga-compensation—two of them, concurrency and temporal budget, addressing classes no ASL tool reasons about. We are deliberate about *which* checks are sound: only the data-flow and structural analyses carry a no-false-positive guarantee, while the semantic checks (typestate, retry, compensation) and the concurrency/temporal checks are conservative, annotation- or heuristic-driven *warnings*. Because raw ASL omits task semantics, STEP CHECK pairs the sound native analyses with a lightweight annotation layer whose defaults are *inferred* from naming and resource conventions, so existing workflows are analysed at zero up-front cost and refined incrementally. Implemented in Rust as a modular frontend/pass

pipeline (4.4k LOC, eight passes), STEPCheck is evaluated on **193** real Step Functions workflows: the sound data-flow analysis fires zero false positives in the wild (corroborating its soundness theorem) and the conservative concurrency analysis likewise fires none; the temporal analysis surfaces 33 unbounded-callback defects; on independently mined fix-commit pairs from public repositories STEPCheck catches **four genuine defects** the commits fixed (missing-field, dead-guard, and unbounded-callback bugs across three real repositories); and a **616**-mutant study detects every injected defect *in the six classes it targets* (**100%** recall on those classes)—where the **statelint** validator catches none of the unsafe retries, missing compensations, concurrency, or temporal defects. Inference recovers persistence at 87% and idempotency at 70% against a human-labelled gold set (two independent annotators, $\kappa = 1.00$), so semantic findings are surfaced as warnings; verification costs $\approx 39 \mu\text{s}$ per workflow; a verified workflow deploys and runs unmodified on AWS Step Functions with zero runtime overhead; and the architecture generalizes to a second workflow language (CNCF Serverless Workflow) with no change to the analyses.

Keywords: Serverless computing · Service orchestration · Workflow orchestration · AWS Step Functions · Static verification · Data-flow analysis · Abstract interpretation · Consumer-driven contracts · Type-state · Saga transactions.

1 Introduction

Cloud-native applications increasingly decompose business logic into independently deployable functions that are coordinated by managed *workflow orchestration* services. Among these, AWS Step Functions has become one of the most widely adopted platforms in industry [26,14]. A Step Functions workflow is a state machine written in the *Amazon States Language* (ASL), a JSON configuration format that describes task invocations, branching, parallelism, retries, and error handling, and that executes on top of AWS Lambda and dozens of service integrations [2].

While orchestration platforms simplify *running* distributed workflows, they provide little support for *reasoning about workflow correctness before deployment*. An ASL state machine is an untyped artifact, and the tooling developers have—**statelint** [3], **asl-validator**, and AWS’s own **ValidateStateMachineDefinition**—validates the JSON schema, the existence of transitions, and the *syntax* of JSONPath expressions (“does this look like a path?”). None of them reasons about whether the field a path *names* can ever have been produced by an upstream state, whether tasks are invoked in a valid business order, whether an automatically retried task is safe to repeat, whether concurrent branches contend on the same resource, or whether a long-running transaction can roll back the effects it has committed. As a consequence, six recurring classes of defect surface only at run time (detailed in Section 2.2): *contract / data-flow mismatches* (a task reads a field no predecessor

produces—the running “`amount` vs. `total`” bug); *protocol/typestate violations* (a task runs out of business order—shipping before payment); *unsafe retries* (a retry re-invokes a non-idempotent operation, duplicating its effect [15]); *concurrency interference* (two `Parallel` branches, or concurrent `Map` iterations, write the same external resource); *temporal-budget faults* (a `waitForTaskToken` callback that can hang indefinitely, or retries that cannot complete within a timeout); and *missing compensation* (a persistent operation cannot be rolled back when a later step fails—an incomplete Saga [12]).

A central obstacle to checking these properties is that ASL *omits the very information the checks require*. ASL has no notion of a task’s input or output type, no business-state labels, no idempotency annotation, and no concept of “compensation” beyond a generic `Catch`. Existing language-level approaches sidestep this by introducing a new programming model with a proprietary runtime [6,27,23]; but the millions of workflows already written in ASL cannot benefit, and re-platforming is rarely an option.

Approach. We present STEP CHECK, a static verifier that runs *before deployment* while remaining fully compatible with AWS Step Functions. Its intellectual centrepiece is a *sound, flow-sensitive data-flow analysis* of the ASL I/O-processing pipeline (`InputPath` $\rightarrow$ `Parameters` $\rightarrow$ `ResultSelector` $\rightarrow$ `ResultPath` $\rightarrow$ `OutputPath`). Cast as an abstract interpretation over a *may-present* lattice of document shapes, it tracks which fields a state’s document can hold and reports a reference to a definitely-absent field—with *no false positives*, a property we establish as a soundness theorem (Section 3.3). This moves the contract check from JSONPath *syntax* (what every existing validator does) to field *provenance*, catching the “`amount` vs. `total`” bug with zero annotations. In service-oriented terms, STEP CHECK is an *orchestration-level integration verifier*: it checks that the data and control contracts between orchestrated services hold *before* deployment, on the platform developers already use. STEP CHECK is organised around two further ideas. First, a *two-layer* model: *native* checks need no input—of these, structural well-formedness and data-flow provenance are *sound* (no false positives), while concurrency-interference and temporal-budget are deliberately *conservative* warnings; *semantic* checks (typestate, retry safety, compensation, typed contracts) consume a lightweight *annotation* layer. Second, because most workflows carry no annotations, the missing semantics are *inferred* from naming and resource conventions, giving a zero-effort default that developers can override. Verified workflows are compiled back to executable ASL, so STEP CHECK adds no runtime component and no lock-in; a typed builder frontend additionally lets greenfield workflows be authored with full static guarantees.

Contributions.

- **A sound data-flow / field-provenance analysis for the Amazon States Language**, formalized as an abstract interpretation over a may-present lattice with a no-false-positive guarantee that we prove (Theorem 1). To our knowledge it is the first *sound, shape-level* data-flow analysis for a

deployed serverless workflow language, and the first static analysis of ASL data flow of any kind beyond JSONPath syntax; it is distinguished from the data-flow-anomaly tradition in business-process modelling (Section 7) by tracking concrete *nested JSON document shapes* through ASL’s exact I/O pipeline—rather than coarse create/read abstractions of a hand-built net—and by running on, and recompiling to, the executable artifact. It has a native tier (no annotations) and a typed tier (declared input schema, a sound static consumer-driven contract) and subsumes the syntactic binding checks (Sections 3.3, 4).

- **Two further analyses for classes no Asl tool addresses:** *concurrency interference* (`Parallel/Map` branches that write the same external resource) and a *retry/timeout-budget* (temporal) analysis (unbounded callbacks, retries that cannot complete within a timeout, heartbeat/timeout mis-configuration).
- A two-layer design that verifies *existing* ASL by pairing sound native checks with an annotation layer whose defaults are inferred from naming/resource conventions, together with the structural, typestate, retry-safety, and compensation checks—eight passes in all (Sections 3–4).
- STEP CHECK, an open, modular Rust implementation (4.4k LOC, eight passes) with pluggable frontends (raw ASL, a typed DSL, the CNCF Serverless Workflow DSL), a diagnostics engine, an ASL emitter, and a mutation-based error-injection tool (Section 5).
- An evaluation on **193 real** workflows (plus 66 CNCF): the data-flow analysis fires *zero* false positives in the wild (soundness), the new analyses surface genuine in-the-wild defects (including **four real bugs** caught in mined fix commits across three public repositories), a **616**-mutant study detects every injected defect *in the six targeted classes* (100% recall on those classes) where `statelint` catches none of the retry, compensation, concurrency, or temporal classes, plus inference accuracy and warning precision against a human-labelled gold set, a typed-tier data-flow recall study, sub-millisecond cost, and a real AWS round-trip (Section 6).

Research questions. **RQ1:** Can a static verifier detect these defect classes in serverless workflows before deployment—and how does it compare to the validator developers already use? **RQ2:** Is the data-flow analysis sound in practice, and can the semantics ASL omits be recovered cheaply enough to analyse existing workflows? **RQ3:** What is the cost of verification and what runtime overhead does it impose? We answer these in Section 6.

2 Background and Motivation

2.1 Step Functions and the Amazon States Language

A Step Functions workflow is a state machine: a `StartAt` entry and a map of named *states*. A `Task` state invokes a Lambda function or a service integration; `Choice` branches; `Parallel` and `Map` fan out; `Wait`, `Pass`, `Succeed`, and `Fail` complete the set. Data flows through a JSON document that each state reshapes with

JSONPath fields (`Parameters`, `InputPath`, `ResultPath`, `OutputPath`), where a key suffixed `.$` denotes a path reference. States carry retry (`Retry`) and error-handling (`Catch`) policies. Listing 1.1 shows a representative task drawn verbatim from our corpus.

Listing 1.1. A real ASL task state (excerpt, `ship-order`).

```
"Get Customer Status": {
  "Type": "Task", "Resource": "arn:aws:states:::lambda:invoke",
  "Parameters": { "FunctionName": "${LambdaGetCustomerStatus}",
    "Payload.$": "$.detail.customer" },
  "Retry": [ { "ErrorEquals": ["Lambda.ServiceException"], "MaxAttempts": 6 } ],
  "Next": "Do Fraud Check"
}
```

The platform validates such a definition’s JSON schema and that every transition target exists, then executes it. It does *not* reason about whether `$.detail.customer` is present, whether retrying this task is safe, or how the workflow recovers if a later step fails.

2.2 Six classes of latent defect

Our running example is the canonical order-processing workflow `CreateOrder` $\rightarrow$ `ReserveInventory` $\rightarrow$ `ChargeCard` $\rightarrow$ `ShipOrder`, which exhibits six defect classes when edited carelessly. (1) *Contract / data-flow*: `ReserveInventory` produces `{orderId, amount}` but `ChargeCard` reads `$.total`—a run-time failure on a missing field. Crucially the path `$.total` is *syntactically* a perfectly good JSONPath, so every schema/syntax validator accepts it; only reasoning about *provenance*—which fields the document can hold at that point—exposes it. (2) *Typestate*: reordering to `ShipOrder` $\rightarrow$ `ChargeCard` ships before payment (valid JSON, invalid protocol). (3) *Retry*: non-idempotent `ChargeCard` with a broad retry (`States.ALL`) may charge twice. (4) *Concurrency*: a `Parallel` that runs `ChargeCard` and a fraud-screen branch which both write the same DynamoDB row, or a `Map` whose concurrent iterations all `PutItem` to one fixed key, races on a shared resource. (5) *Temporal*: a `waitForTaskToken` approval step with no `TimeoutSeconds/HeartbeatSeconds` blocks the execution for up to a year if the token is lost. (6) *Compensation*: `ReserveInventory` commits a reservation that leaks if `ChargeCard` fails with no release path. None is caught by the platform; all six are recognised, under-tooled failure modes in cloud-native systems [15,14]. We claim nothing about their prevalence—curated examples are mostly correct (Section 6.2)—only that they escape static checking entirely today; Section 6 measures detection on both unmodified and injected defects.

2.3 The information gap

The obstacle to checking these properties statically is not the analysis but the *input*. Raw ASL encodes control flow and JSONPath data plumbing, but it does

not record: the *type* of data a task consumes or produces; the *business state* a task represents; whether a task is *idempotent*; or whether a task is *persistent* and which action *compensates* it. A verifier for existing workflows must therefore either (a) restrict itself to properties decidable from ASL alone, or (b) recover the missing semantics. STEP CHECK does both: it runs sound native checks that need nothing beyond ASL, and it recovers the remaining semantics through a lightweight annotation layer whose defaults are inferred from the strong naming and resource conventions that pervade real workflows (Section 3).

3 Approach

3.1 A two-layer analysis model

STEP CHECK separates what can be checked from ASL alone from what needs recovered semantics. *Native* analyses (Layer A) require no annotations and divide by confidence. *Two are sound* (no false positives): structural well-formedness, and the *data-flow provenance* analysis of Section 3.3 (which subsumes the ASL-semantic binding checks—a `waitForTaskToken` task that never passes its token, `$.Map` misuse, invalid JSONPath payloads—and adds field-availability reasoning). *Two more are native but deliberately conservative warnings*: the *concurrency-interference* check (`Parallel/Map` branches that write a shared resource) and the *retry/timeout-budget* check (unbounded callbacks, retries that overrun a timeout, heartbeat/timeout mis-configuration), which under-approximate to avoid false alarms rather than claiming soundness. *Semantic* analyses (Layer B)—typestate, retry safety, compensation, and typed contract compatibility—consume an *annotation* layer that records the task semantics ASL omits. (Concurrency and retry safety also *read* the annotation layer to learn which tasks write or are non-idempotent, so they sharpen as annotations are added, but their structural triggers fire natively.) The annotation layer has two sources, in priority order: explicit annotations (authored by hand or produced by the typed DSL frontend), and *inference* that supplies defaults from naming and resource conventions. Findings derived from inferred facts are reported at *warning* severity, while findings derived from declared facts are *errors*; this single rule keeps confidence visible throughout the tool’s output. We are deliberate about what “verification” means at each layer: the *sound* native checks (structural well-formedness and data-flow provenance) need no input and carry a no-false-positive guarantee; the conservative native checks (concurrency, temporal budget) raise heuristic warnings; whereas the semantic checks are only as strong as the semantics they are given—with declared annotations they are precise errors, and in the zero-annotation default they degrade gracefully to best-effort, inference-driven *warnings* (typestate, having no labels to act on, simply stays silent). STEP CHECK is thus a sound linter that becomes a precise verifier as annotations are added.

3.2 Workflow model

We model a workflow as a graph of states with a designated start. Beyond its control-flow successors, a *task* carries four pieces of semantic information.

Contract: a task $T : A \rightarrow B$ consumes/produces record types (field sets); adjacent $T_1 : A \rightarrow B$, $T_2 : C \rightarrow D$ *compose* only if B provides every field C requires (natively, without declared types, the layer instead checks that JSON-Path references resolve to data the workflow binds). **Typestate:** T advances a *business state* $s_{\text{in}} \rightarrow s_{\text{out}}$; a workflow is well-ordered when each task consumes the state its predecessor produced, and an optional *protocol* (e.g. *created* $\rightarrow$ *reserved* $\rightarrow$ *paid* $\rightarrow$ *shipped*) lists permitted transitions. **Idempotency:** repeating an idempotent task causes no extra external effect, so it may be retried freely; a non-idempotent one may not be retried on broad error classes. **Persistence/compensation:** a *persistent* task commits a durable resource (reservation, charge, created record) that a later failure should undo, so it needs a *compensation*—an explicit handler or a **Catch** to a compensating state. This vocabulary is deliberately lightweight—no separate formal calculus—yet exactly what the semantic checks of Section 4 consume.

3.3 Sound data-flow provenance by abstract interpretation

The contract defect of Section 2.2 is a *data-flow* fact: `$.total` fails not because it is malformed—it is a valid path—but because no execution reaching **ChargeCard** produces a `total` field. Deciding this requires modelling how ASL reshapes its state document, which it does through a fixed five-stage pipeline per state: **InputPath** selects a sub-document; **Parameters/ItemSelector** *construct* the task input (each `"k.$":"$.p"` entry *reads* `$.p`); the task result is combined through **ResultSelector** and merged into the *raw* input at **ResultPath**; and **OutputPath** selects the state output. We give this an abstract interpretation.

Domain. A document shape over-approximates the set of field paths that *may* be present:

$$\sigma ::= \top \mid \{k_1:\sigma_1, \dots, k_n:\sigma_n\}$$

where $\top$ is opaque (any field may be present—e.g. an arbitrary Lambda result or the execution input) and a record lists *exactly* the keys that may occur (a key absent from the record is absent on every represented document; scalars/arrays are the empty record / $\top$). The order is $\sigma \sqsubseteq \top$ and pointwise on records; the join $\sqcup$ is the *union* of keys ($\top$ absorbs), so a key present on any incoming path survives. A path `$.p1...pk` is *definitely absent* in σ iff descending the records hits a missing key without first reaching $\top$.

Transfer functions. Each stage is a monotone function on shapes: **InputPath/OutputPath** re-root by descent (a missing or non-trivial selection yields $\top$); **Parameters/ItemSelector/ResultSelector** build a record whose keys are the constructor’s keys (a `.$` leaf reads an opaque value, hence $\top$); the opaque task result is $\top$ unless a **ResultSelector** pins its keys; **ResultPath** places the result into the raw input, preserving siblings (`$` replaces, `null` discards). A state in *JSONata* mode (or one using the JSONata **Arguments/Output** constructs) reshapes its document with expressions outside our JSONPath

fragment, so it yields $\top$ and its references are not checked—trading precision for soundness on JSONata workflows. A forward fixpoint over the control-flow graph—joining at merge points and along **Catch** edges (which carry an opaque error object), and converging in a number of rounds bounded by the machine’s finite key universe (Theorem 1)—computes an entry shape for every state. We then check exactly the references the ASL runtime *evaluates and aborts on when absent*: the ‘\$’ constructors of **Parameters/ItemSelector** and a **Map’s ItemsPath**. We deliberately do *not* check **Choice** operands: a **Choice** comparison (**IsPresent**, **IsNull**, value comparators) is a *test* that returns false on an absent field rather than failing the execution—flagging it would be unsound.

Soundness. Let $\gamma(\sigma)$ denote the set of JSON documents a shape σ admits: $\gamma(\top)$ is every document, and $\gamma(\text{Obj } r)$ is the documents whose top-level keys lie within $\text{dom}(r)$, each value v at key k lying in $\gamma(r(k))$. The analysis reports SC1101 for a reference only when its path is *definitely absent* in the state’s entry shape. Each transfer over-approximates, which gives the central guarantee.

Theorem 1 (No false positives). *If STEP CHECK reports SC1101 for a path p at a state s , then in every concrete execution that reaches s , p is absent from the state document, so the ASL runtime fails the reference. Equivalently, the analysis never flags a reference some execution could satisfy.*

Proof (sketch). Every stage transfer $f^\#$ is *sound* for γ , i.e. $f(d) \in \gamma(f^\#(\sigma))$ for all $d \in \gamma(\sigma)$: a **.\$** constructor binds its key to $\top$; the opaque task result and any JSONata-mode state yield $\top$; **ResultPath** writes that result while preserving siblings; **InputPath/OutputPath** descend. The join is the key-union, sound for γ since $\gamma(\sigma_1) \cup \gamma(\sigma_2) \subseteq \gamma(\sigma_1 \sqcup \sigma_2)$. By induction over the control-flow graph—taking the join over all incoming normal *and* **Catch** edges—the computed entry shape σ_s satisfies $d \in \gamma(\sigma_s)$ for every document d of every execution reaching s . If SC1101 fires, p descends σ_s to a missing key without meeting $\top$, so by the definition of γ no $d \in \gamma(\sigma_s)$ contains p ; hence p is absent on every reaching execution. *Termination:* shapes form an ascending chain (a join only adds keys or lifts to $\top$) in a lattice whose height is bounded by the machine’s finite key universe κ (keys occur only in the finitely many constructor/result literals and the declared input), so the iteration reaches a least fixpoint within $|S| \cdot (\kappa + 2)$ rounds for $|S|$ states; the implementation enforces this bound and reports only at the fixpoint.

The guarantee is thus established, not merely asserted; we additionally corroborate it empirically—zero SC1101 on 193 unmodified workflows, and an independent per-path *execution oracle* (a second interpreter that shares no code with the analysis) confirms all 84 injected-defect findings absent on every reaching path, with zero counterexamples (Section 6.5). The analysis has two tiers that mirror the tool’s two layers. The **native** tier seeds the start document as $\top$ (the execution input is arbitrary), so findings arise only where the workflow itself constructs the data it later reads—a **Pass Result**, a **ResultSelector**, a static **Parameters** object, or an **OutputPath** that drops a field; the statement above holds *unconditionally* for this tier. The **typed** tier seeds the start as a closed record of the declared input

schema (sidecar, DSL, or CNCF JSON Schema), which recovers the full “field never produced” class flow-sensitively through the `ResultPath` merge—strictly more than an adjacent-task field-subset check. There the guarantee is relative to a *closed-world reading* of the declared contract: the schema enumerates the fields the input may contain. This matches the intent of an author who declares a contract; where inputs are genuinely open, the native tier (which assumes nothing) is the sound fallback.

3.4 The annotation layer

Explicit semantics are supplied either through a *sidecar* file attached to an existing ASL workflow (Listing 1.2) or, for greenfield workflows, through a typed builder DSL that emits both the ASL and its annotations (one `.task(...)` call per state carries its schema, protocol edge, idempotency, persistence, and compensation). The two routes resolve to the same internal representation, so the analyses are agnostic to how a workflow was authored.

Listing 1.2. A sidecar annotation (TOML), excerpt.

```
[schemas.OrderReserved]
fields = ["orderId", "amount", "reservationId"]
[tasks.ChargeCard]
idempotent = false
persistent = true
compensation = "RefundPayment"
state_in = "OrderReserved"
state_out = "OrderPaid"
```

3.5 Inference of default annotations

Most existing workflows carry no annotations, yet their task names and resource ARNs are highly conventional. STEP CHECK exploits this with a documented, configurable heuristic that classifies each task from a lowercased *signal* (its name, its `Resource`, and parameters such as `FunctionName`). The heuristic uses four small keyword sets: *read* verbs (`get`, `list`, `describe`, `query`, `validate`, ...) imply idempotent and non-persistent; *write* verbs (`create`, `charge`, `send`, ...) imply non-idempotent; *resource-acquiring* verbs (`reserve`, `pay`, `provision`, `putItem`, ...) additionally imply persistent; and *notification* verbs/resources (`notify`, `publish`, `:sns:`, ...) are non-idempotent but, being fire-and-forget, never persistent. When no keyword matches, the heuristic *abstains*, leaving the task unclassified rather than guessing. Inference thus yields a zero-effort default whose mistakes (Section 6) are visible (warnings) and correctable (an explicit annotation always wins).

3.6 Architecture

The pipeline is a straight line: a *frontend* lowers a surface syntax to a format-agnostic intermediate representation (IR); annotation resolution attaches declared

Table 1. Diagnostic codes emitted by STEP CHECK. Severity $\dagger$ is *error* when derived from declared facts and *warning* when derived from inference.

Code	Pass	Meaning
SC0001–0010	structural	undefined start/transition target, unreachable state, dead-end, non-exhaustive Choice , empty Map/Parallel , malformed state
SC1001–1003	data-flow	waitForTaskToken without a task token, \$\$. Map misuse, invalid JSONPath payload
SC1010	data-flow	typed contract mismatch: successor requires a field the predecessor does not produce
SC1101	data-flow	reference to a field no execution can have produced (provenance; sound)
SC1110	data-flow	dead Choice guard: a branch tests a field provenance proves absent here (path-sensitive; sound)
SC2001–2002	typestate	out-of-order composition; undeclared business-protocol transition
SC3001 $\dagger$	retry	non-idempotent task retried on a broad error class
SC4001–4002 $\dagger$	compensation	persistent task with no compensation/handling; compensation names a missing state
SC4010 $\dagger$	compensation	persistent task whose only error path never compensates (effect-aware Catch)
SC5001–5002	concurrency	Parallel branches / concurrent Map iterations write the same resource
SC6001–6003	temporal	unbounded callback; retry budget exceeds machine timeout; heartbeat $\geq$ timeout

and inferred semantics to the IR; a *pass* pipeline runs the eight analyses, each emitting coded diagnostics; and an *emitter* lowers verified workflows back to ASL. The design is modular along two axes: a new input format is a new **Frontend**, and a new check is a new **Pass** (Section 5)—indeed the three analyses new to this paper were each added as a single **Pass** with no change to the IR or the other checks.

4 Static Analyses

STEP CHECK runs eight independent passes over the IR, each emitting diagnostics with a stable code (Table 1), a severity, the offending state, and a remediation note. All are linear in the number of states (the data-flow pass, linear modulo the control-flow fixpoint) and recurse into nested **Map/Parallel** sub-machines.

Structural validation (SC0xxx). A reachability traversal from **StartAt** (following normal and error transitions) checks that all transition targets exist, that no state is unreachable, that every non-terminal offers a continuation, that a terminal is reachable, that **Choice** states declare a **Default**, and that **Map/Parallel** states are well-formed—sound checks that catch defects AWS reports only at run time.

Data-flow provenance (SC1xxx, SC1101). This is the analysis of Section 3.3, run as a forward abstract interpretation that computes each state’s document shape and checks the data references a state evaluates against it (its `Parameters/ItemSelector` ‘\$’ constructors and `ItemsPath`). A reference to a path that is *definitely absent* in the state’s shape is SC1101—a sound finding (no false positives). Three legacy ASL-binding checks are special cases the same pass still reports: a `waitForTaskToken` task must pass `$$Task.Token` (SC1001); the `$$Map` context is valid only inside a `Map` (SC1002); and every payload must be a valid path or `States.*` intrinsic (SC1003). SC1101 goes strictly beyond all of these and beyond what `statelint/asl-validator` check: it is the only one that asks whether the field a *valid* path names can ever have been produced. With a declared input schema the shape is closed and the analysis recovers the full `amount/total` class (Section 2.2); STEP CHECK additionally retains an adjacent-task field-subset check (SC1010) as a cheap complement when only per-task schemas, not a workflow input schema, are declared. The same shapes support a *path-sensitive* finding, SC1110: a `Choice` branch whose match requires a field that is definitely absent in the state’s shape (e.g. `IsPresent` or a value comparator on a never-produced field) can never be taken, so the branch is dead. It is sound for the same reason as SC1101—we flag only on a definitely-absent operand, never on `Top`—and is the first step of refining the data-flow lattice with the information `Choice` predicates carry.

Typestate (SC2xxx). Using `state_in/state_out` labels, the pass checks that adjacent tasks compose (predecessor output state = successor input state, SC2001), catching re-orderings such as ship-before-pay, and that each task’s transition is a permitted protocol edge when one is declared (SC2002). Absent annotations it is a no-op.

Retry safety (SC3xxx). A task with a `Retry` policy whose `ErrorEquals` includes a broad class (`States.ALL/States.TaskFailed`) is flagged SC3001 if its idempotency resolves to *non-idempotent*. Retries restricted to transient errors (throttling, `ServiceException`) are not flagged, as they fire before any effect.

Compensation completeness (SC4xxx). A *persistent* task needs either a declared compensation or a `Catch` path; one with neither leaves its effect uncompensated (SC4001), and a compensation naming a non-existent state is SC4002. A bare “has a `Catch`” rule over-accepts, however: a `Catch` to a logger or a `Fail` does not undo the effect. The *effect-aware* check SC4010 closes this gap—it walks each `Catch` target’s reachable tasks and flags a persistent task whose only error path never reaches a compensating action (*refund/cancel/release/rollback...*). Unlike SC1101/SC5xxx, this fires in the wild (Section 6.2); it is a warning, since the classification is name-based.

Concurrency interference (SC5xxx). `Parallel` branches and `Map` iterations each receive their own copy of the state document, so the document cannot race—but the *external* resources their tasks touch can. The pass mines each task’s

resource (its integration service plus the named target: `TableName`, `QueueUrl`, `TopicArn`, `Bucket`) and, to avoid false positives on idiomatic keyed/commuting patterns, considers only *overwriting* writes—`PutItem`/`PutObject`, which are last-writer-wins. It excludes `UpdateItem` (an atomic merge that commutes for disjoint attributes) and fire-and-forget `Publish`/`SendMessage` (independent messages). It then flags two patterns: two `Parallel` branches that both overwrite the *same* resource (SC5001); and a concurrent `Map` (`MaxConcurrency` $\neq 1$) whose iterations overwrite a shared resource while threading *no* per-item data (no ‘\$’ reference in the write, so every iteration clobbers the same item, SC5002). Both are *warnings*, consistent with the tool’s confidence model. No prior ASL tool reasons about cross-branch interference.

Retry / timeout budget (SC6xxx). ASL’s resilience knobs interact in ways the platform never cross-checks. A `waitForTaskToken` (or `Activity`) task with neither `TimeoutSeconds` nor `HeartbeatSeconds` waits up to a year for a token that may never arrive (SC6001, the most common in-the-wild finding—human-approval and external-callback steps). When the state machine declares a `TimeoutSeconds`, the pass computes a task’s worst-case retry budget as the geometric back-off sum $\sum_k \text{interval} \cdot \text{backoff}^k$ and flags retries that cannot complete within it (SC6002). Finally `HeartbeatSeconds` $\geq$ `TimeoutSeconds` is a definite mis-configuration the platform rejects only at deploy time (SC6003).

Worked example. `STEPCHECK` reports no findings on the well-formed order workflow. Reordering it to ship before charging (Section 2.2) yields five errors and no false positives: two SC2001 typestate mismatches (e.g. `ShipOrder` consumes `OrderPaid` but its new predecessor produced `OrderReserved`) and three SC1010 contract mismatches (`paymentId`, `amount`, `reservationId` required but not produced upstream)—complementary checks: typestate explains the ordering fault, the contract check pinpoints the broken data. The native data-flow tier needs no annotations at all: on a workflow that constructs `{order:{orderId,amount}}` in a `Pass` and then reads `$.order.total`, `STEPCHECK` reports SC1101 directly—the `amount/total` bug caught on raw ASL, which `statelint` accepts as a valid path.

5 Implementation

`STEPCHECK` is implemented in Rust as a single command-line tool, `stepcheck` (4,394 lines). We chose Rust deliberately: a standalone analyzer with a `serde`-based parser is what makes it possible to ingest the *existing* ASL JSON found in real repositories—rather than only workflows authored in a new embedded language—and it yields a single dependency-light binary with predictable, sub-millisecond performance.

5.1 A modular frontend/pass pipeline

Extensibility rests on two traits (Listing 1.3): a `Frontend` lowers a surface syntax to the IR, a `Pass` reads only the IR and writes diagnostics. Adding an input

format is a new **Frontend**; a check, a new **Pass**. We exercised this directly—the tool ships three frontends (typed DSL, raw ASL, CNCF [7]), the last added *without changing the IR or any pass* (Section 6.10).

Listing 1.3. The two extension points.

```
pub trait Frontend { fn load(&self, src: &str) -> Result<Workflow>; }

pub trait Pass {
  fn id(&self) -> &'static str;
  fn run(&self, wf: &Workflow, sink: &mut DiagnosticSink);
}
```

5.2 Components

The **workflow IR** is a format-agnostic graph of states (**Task**, **Choice**, **Parallel**, **Map**, **Wait**, **Pass**, **Succeed**, **Fail**) with control flow, JSONPath data-flow fields, retry/catch policies, nested sub-machines, and a slot for resolved annotations. The **Asl frontend** lowers from a generic `serde_json` value rather than rigid structs—so the diversity of real ASL (intrinsic, `.$` parameters, the `$$` context, SAM `${}` placeholders, classic **Iterator** vs. distributed **ItemProcessor**, JSONata) never causes a hard failure—and parses all 193 corpus workflows without error. **Annotation resolution** loads a TOML sidecar, resolves schema names to field sets, and fills gaps with the heuristic of Section 3.5. The **passes and diagnostics** implement the eight analyses plus an engine that renders findings for a terminal or as JSON; the data-flow pass is the largest at ≈ 400 lines (the shape lattice, transfer functions, and control-flow fixpoint), the rest a few dozen lines each. The **Asl emitter** lowers a verified IR back to deployable ASL JSON; it is the DSL frontend’s code-generator and the basis of the AWS round-trip (Section 6)—deploying its output surfaced (and we fixed) a real bug where an explicit **End** was emitted on **Succeed/Fail** states, which Step Functions rejects. The emitter faithfully reproduces control flow, retries, catches, and per-state timeouts; it is currently *lossy* for a workflow’s pure I/O-processing fields (it does not re-emit **InputPath/OutputPath/ResultSelector** or the machine-level **TimeoutSeconds**). The round-trip deploys a DSL-authored workflow that uses none of those fields, so the deployed machine is behaviourally equivalent; full-fidelity emission for arbitrary ingested ASL is a small extension we note rather than claim.

Execution oracle. An independent *execution oracle* (`concrete.rs`) re-interprets a workflow concretely—enumerating the document along each acyclic path, with opaque task results modelled explicitly—and cross-checks every data-flow finding against this second, code-disjoint implementation (Section 6.5). It witnesses the soundness theorem by agreement between two independent algorithms.

CLI and error injection. `stepcheck` exposes `check`, `infer`, `scan`, `stats`, `eval`, `oracle`, `eval-pairs`, `mutate`, and `demo` (the last emits the built-in typed-DSL

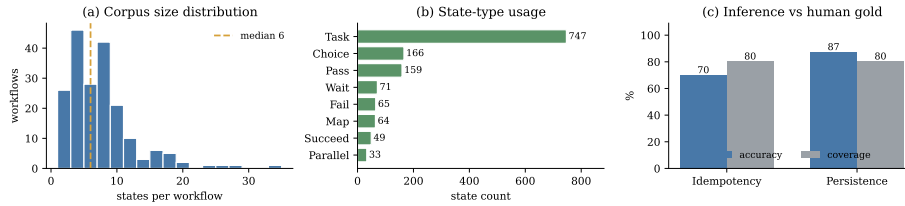


Fig. 1. Evaluation overview. (a) Distribution of workflow sizes and (b) state-type usage across the 193-workflow corpus; (c) inference accuracy and coverage for idempotency and persistence against the human-labelled gold set (Section 6.7).

example to ASL). The `eval-pairs` command drives the real-bug benchmark (prefix/post-fix replay); `oracle` runs the soundness cross-check above. The `mutate` command injects exactly one defect of a chosen class into a workflow—a broken JSONPath binding, a reference redirected to an absent field (data-flow), a broad retry on a non-idempotent task, a removed compensation, a dangling transition, two `Parallel` branches made to write one resource, or a heartbeat $\geq$ timeout—and re-emits deployable ASL; it drives the mutation study of Section 6.

The **4,394** lines divide as: workflow IR 323; three frontends (ASL parse/emit, typed DSL, CNCF) 766; annotations + inference 275; the eight analysis passes + manager 1,615; diagnostics 115; the execution oracle 269; mutation engine 316; and CLI + statistics + evaluation + oracle harness 715.

6 Evaluation

We evaluate STEP CHECK on real serverless workflows, asking: can it detect the six defect classes before deployment (RQ1); can it recover the semantics ASL omits cheaply enough to analyse existing workflows (RQ2); and what does verification cost, at build and run time (RQ3)? Every number below is produced by the released tool over a committed corpus and is reproducible with the commands shown.

6.1 Setup and corpus (RQ1)

We mined **193** Step Functions workflows from two public AWS repositories (`aws-samples/aws-stepfunctions-examples` and `step-functions-workflows-collection`), taking every file that parses as an ASL state machine. The corpus (Fig. 1) is realistic and diverse: 1,355 states, median 6 per workflow (max 33), and broad use of `Choice` (166), `Map` (64), `Parallel` (33), retries (84 workflows), and catch handlers (51). All experiments ran on a commodity laptop (Rust 1.96 release build).

6.2 Findings in the wild (RQ1)

Run unmodified over the corpus (with inference enabled), STEP CHECK flags **95 of 193** workflows. We separate *sound native* findings, which need no annotations, from *inference-driven* findings, which are warnings requiring triage. Among the native findings, nine workflows contain a **Choice** with no **Default** (SC0007)—a recognised cause of **States.NoChoiceMatched** run-time failures—and one is genuinely malformed (SC0010): an aws-samples example nests a top-level **QueryLanguage** directive *inside* its **States** block, where Step Functions would reject it. The temporal analysis adds **33** findings, all SC6001: unbounded **waitForTaskToken** callbacks—human-approval and payment/shipping steps with no timeout or heartbeat, a real anti-pattern AWS documents against. The data-flow (SC1101, SC1110) and concurrency (SC5xxx) analyses fire on *none* of the 193: SC1101 because the analysis is sound and these samples are correct (Section 6.5), SC1110 because no corpus workflow guards a **Choice** on a field it never produces, and SC5xxx because its conservative overwriting-write rule finds no genuine last-writer-wins race among the corpus’s (correctly keyed or commuting) concurrent writes—all precision results, with detection power shown by the mutation study below. The effect-aware compensation check *does* surface a new class in the wild: SC4010 flags **28** persistent tasks whose only **Catch** merely logs, notifies, or fails—never reaching an undo action—a finding a bare “has-a-Catch” rule misses entirely. The inference-driven warnings (SC3001: 36; SC4001: 117; SC4010: 28) mark non-idempotent tasks retried broadly and persistent tasks whose failure paths do not (or cannot) compensate; as Section 6.7 shows, the SC3001/SC4001 class has measured precision of 60% on gold-covered findings (70% for the compensation warnings SC4001, 100% for SC4010) but includes false positives (e.g. a **RefundPayment** compensator inferred as needing its own compensation), which is precisely why they are surfaced as warnings rather than errors.

6.3 Mutation-based detection (RQ1)

Because correct real workflows expose few of the deeper defects, we measure detection power by *error injection*. For each defect class we inject exactly one defect per applicable workflow with **stepcheck mutate** and re-verify; a defect counts as *detected* only if the expected diagnostic’s count strictly increases over the unmutated baseline (so we measure detection of the *injected* defect, not pre-existing findings). Across **616** mutants spanning six classes—the four above plus a shared-resource **Parallel** write (SC5001) and a heartbeat $\geq$ timeout (SC6003)—STEP CHECK achieves **100% recall** (Table 2, STEP CHECK column), and it never reports the expected defect on the corresponding originals. (The data-flow class is studied separately in Section 6.5 because, unlike the others, its detection depends on a known document shape rather than firing on raw ASL.) We stress that this measures *reliability of each check on the defect class it targets*, not detection of arbitrary bugs; the mutations are constructed to fall within each check’s remit. To show the checks are *specific* and not merely trigger-happy, the harness also emits a per-operator confusion matrix: it is near-diagonal—each

semantic check fires only on its own class (the retry mutant trips SC3001 and nothing else; the concurrency mutant only SC5001)—and the only off-diagonal entries are genuine structural *consequences* of an edit (redirecting a transition also orphans its old target, SC0003), not misfires of unrelated checks. To complement this synthetic study with *independently*-introduced defects, we also mine real fix-commit pairs and replay STEP CHECK on them (`stepcheck eval-pairs`); this finds four genuine in-the-wild defects, reported in Section 6.6. The next subsection first contrasts the mutation results with what a standard validator detects on the same mutants.

6.4 Comparison to a standard validator (RQ1)

The natural baseline is the validation a Step Functions developer already has. `statelint` [3] (the AWS Labs reference linter; we ran v0.8.0) checks an ASL definition against the J2119/ASL schema plus structural rules. We ran it on the same corpus and the same 616 mutants, scoring a mutant *detected* only when statelint emits a new problem that names the injected fault (Table 2). The two tools *agree* on the defects statelint can express: it flags every dangling transition (166/166) and—because a broken JSONPath binding violates the schema—almost every broken *binding* (140/141). But it detects **0** unsafe retries, **0** missing compensations, **0** concurrency-interference defects, and **0** temporal-budget defects: a broad `States.ALL` retry, an absent `Catch`, two branches writing one table, and a heartbeat that exceeds a timeout are all schema-valid, so a validator without a notion of idempotency, persistence, resources, or a temporal budget cannot see them. (A naive count-of-problems metric would appear to “detect” many of these—the injected retry’s float `IntervalSeconds` once tripped statelint’s integer pedantry, a removed `Catch` orphans its handler state—but none of those complaints is the analysis in question, so we do not credit them.) In the wild the asymmetry is starker still: statelint flags 113/193 workflows with 499 problems, **410 of them** schema-shape nits (e.g. `BackoffRate/IntervalSeconds` number typing) and none touching the six semantic classes, whereas STEP CHECK’s findings are all semantically meaningful. STEP CHECK is thus complementary to and strictly more discriminating than schema validation on the properties that motivate it.

6.5 Data-flow provenance: soundness, power, and reach (RQ1/RQ2)

The data-flow analysis (Section 3.3) is the paper’s central new check, and its claim is *soundness*—no false positives—so we evaluate it on its own terms. *Precision*. Run over all 193 unmodified workflows, SC1101 fires **zero** times. These are mostly correct sample workflows, and the analysis is conservative by construction; firing on none of them is the expected, desired outcome and empirically corroborates the soundness statement of Section 3.3. *Independent oracle*. To validate soundness beyond the absence of findings, we cross-check with a per-path *execution oracle* (`stepcheck oracle`): a second interpreter that shares no code with the analysis, enumerates the concrete document along every acyclic path

Table 2. Injected-defect detection over 616 mutants (delta over baseline): STEPCheck vs. `statelint`. A mutant counts as detected only when the tool emits a new problem naming the injected fault.

Defect class	Check	Applicable	STEPCheck	<code>statelint</code>
Dangling transition	SC0002	166	166 (100%)	166 (100%)
Broken data binding	SC1003	141	141 (100%)	140 (99%)
Unsafe retry	SC3001	103	103 (100%)	0 (0%)
Missing compensation	SC4001	19	19 (100%)	0 (0%)
Concurrency interference	SC5001	21	21 (100%)	0 (0%)
Temporal (heartbeat/timeout)	SC6003	166	166 (100%)	0 (0%)
Total		616	616 (100%)	306 (50%)

to a state, and confirms a flagged field is genuinely absent on all of them. Over the **84** injected missing-field defects the typed tier detects (below), the oracle confirms *every one* absent on all reaching paths with **zero** counterexamples—agreement between two independent algorithms, which is strong evidence the transfer functions and join are sound. *Power.* That the check is not vacuous we show on worked demonstrators (released in the corpus). On a *native* example—no annotations—where a `Pass` builds `{order:{orderId,amount}}` and a later task reads `$.order.total`, STEPCheck reports SC1101 on raw ASL; `statelint` and `asl-validator` accept it (the path is valid). On a *typed* example with a three-field input schema, it reports two SC1101: a schema miss (`$.total` for `amount`) and a *flow-sensitive* miss—`$.reservation.reservationCode` where an upstream `ResultSelector/ResultPath` merge produced `reservationId`—a defect an adjacent-task field-subset check cannot see because the field is created two states earlier. *Reach.* To measure detection at corpus scale we run the typed tier: each workflow’s start document is seeded as a closed record of the top-level fields it references, and we inject a missing-field defect per workflow (redirecting a reference to an absent field). STEPCheck detects **84 of 126 (67%)**; the undetected third are references that sit *downstream of an opaque task result*, where the document shape has soundly collapsed to $\top$ and no missing-field claim is possible. This number is therefore not a reliability gap but a measurement of ASL’s intrinsic opacity: roughly two-thirds of field references on real workflows are statically recoverable, and the remainder are exactly what richer typed outputs (a `ResultSelector`, or CNCF’s declared output schemas, Section 6.10) push back. The baseline produces no spurious SC1101 under the closed-world seed either, so the typed tier inherits the native tier’s precision.

6.6 Real defects in the wild (RQ1)

The mutation study measures reliability on *injected* defects; to complement it with *independently*-introduced ones, we mined fix-commit pairs from public Step Functions repositories (`eval/mine_realbugs.js`: commits whose message looks like a fix and whose diff edits an ASL definition; we extract the file *before* and *after*

the commit) and replayed STEP CHECK on each with `stepcheck eval-pairs`. On 4 of 58 mined pairs STEP CHECK reports a defect on the pre-fix version that the fix removes, and we manually confirmed *all four* as genuine defects in its targeted classes, across three independent repositories (one an official AWS sample):

- **SC1101** (`aws-batch-runtime-monitoring`): a task reads `$.Execution.Input`—a single-`$` path selecting a *document* field no state produces, where the author meant the context object `$$Execution.Input`; the fix routes the value through a real field.
- **SC1101** (`serverless-ai-fitness`): two tasks read `$.profile.Item/$.userId`, fields not produced at those states; the fix refactors the user-data flow onto produced fields.
- **SC1110** (`serverless-ai-fitness`): a Choice guards on `$.profile.subscription.level`, a field never produced—so the subscriber branch is unreachable and subscribers were never routed; the fix corrects the path. Our path-sensitive dead-guard check thus finds a real *logic* bug.
- **SC6001** (`leave-management`): a callback task has no `TimeoutSeconds/HeartbeatSeconds` and can hang indefinitely; the fix adds `TimeoutSeconds`.

On the remaining 54 pairs—predominantly value/config corrections (a bucket name, an API endpoint, a parameter) outside STEP CHECK’s remit—it reports no fix-removed defect, and on every pre/post pair the flagged code is *absent* from the post-fix version (no residual false positive for the fix). Small as it is, this is a genuine in-the-wild result on the *deployed* artifact: the centrepiece provenance analysis, the new dead-guard check, and the temporal check each catch a real, independently-fixed defect. We release the miner and the adjudicated benchmark (`corpus/realbugs/`) so it can be grown.

6.7 Inference accuracy (RQ2)

The semantic checks depend on recovering idempotency and persistence, so we measure the heuristic against a *human* gold standard of **147 tasks across 23 workflows**—spanning Sagas, an inventory domain, account vending, security automation, media/AI pipelines, CQRS, data migration, and more—labelled by each task’s true operational semantics (the annotators confidently label 137; the rest are abstentions). To guard against labeller bias, the 13 workflows added in this revision were labelled by *two independent human annotators* under a written rubric and reconciled; agreement was perfect (Cohen’s $\kappa = 1.00$ for *both* idempotency and persistence over the 116 doubly-labelled tasks), and the original 10 were hand-labelled by an author. We release every label, the rubric, and the scoring harness for scrutiny. The heuristic abstains on $\sim 20\%$ of tasks (unconventional verbs such as *confirm*, *audit*, *aggregate*); on the rest it reaches **87.3%** accuracy for persistence but only **70.0%** for idempotency (Table 3). The asymmetry is the headline result of this experiment, and it is honest: persistence (which drives the compensation check) tracks resource-acquiring verbs

Table 3. Inference accuracy vs. the human-labelled gold set (23 workflows; two independent annotators on the 13 added workflows, Cohen’s $\kappa = 1.00$; “Labelled” is tasks with a confident gold label; accuracy is over non-abstained predictions).

Property	Labelled	Predicted	Abstained	Correct	Accuracy	Coverage
Idempotency	137	110	27	77	70.0%	80.3%
Persistence	137	110	27	96	87.3%	80.3%

well, whereas idempotency is genuinely hard from names alone—*cancel/release* compensators and stable-key writes (a DynamoDB `PutItem/UpdateItem` keyed by a primary key) are idempotent *in effect* but read as writes. That perfect inter-annotator agreement ($\kappa = 1.00$) coexists with imperfect heuristic accuracy means the truth is clear to a human but the names do not carry it. This is precisely why inferred findings are *warnings*: a one-line annotation overrides any default, and the residual error is visible, not silent.

Precision of in-the-wild warnings. The gold set lets us measure the precision of the inference-driven warnings of Section 6.2 directly: for each warning that lands on a gold-labelled task, a finding is a true positive iff the gold label confirms it. Over the 30 gold-covered SC3001/SC4001 findings, **18 (60%)** are true positives—**16/23 (70%)** for the compensation warnings SC4001, but only **2/7 (29%)** for the broad-retry warnings SC3001, mirroring the weaker idempotency inference that drives them; the new effect-aware SC4010 is **6/6 (100%)** on its gold-covered findings. Every false positive is a compensator (`RefundPayment`, `CancelFlightReservation`) or an idempotent keyed/metadata write—the same patterns above—which is exactly why these are surfaced for triage rather than as errors. RQ2 answer: persistence is recovered well and the compensation warnings are mostly actionable; idempotency-driven retry warnings are noisier and benefit most from a declared annotation.

6.8 Verification cost and runtime overhead (RQ3)

Verification is cheap: the eight passes—including the data-flow fixpoint—average $\approx 39 \mu\text{s}$ /workflow (median $\approx 25 \mu\text{s}$, max $\approx 1.1 \text{ ms}$); the whole corpus verifies in $\approx 7.5 \text{ ms}$. The three new analyses roughly doubled the per-workflow cost over the five-pass version yet kept it firmly sub-millisecond on all but one workflow. Because all analysis is ahead-of-deployment and the output is ordinary ASL, the *runtime* overhead is **zero**—no middleware, sidecar, or instrumentation—unlike runtime fault-tolerance and transaction approaches [27,6].

6.9 Conciseness and AWS round-trip

The typed DSL is also concise: 20 declarative statements expand to a 119-line ASL definition while additionally encoding five schemas, a four-edge protocol, and per-task idempotency/persistence/compensation that ASL cannot express. To

confirm compatibility end-to-end, we deployed the DSL-verified order workflow to real AWS Step Functions *unmodified* (only binding `FunctionName` to a Lambda ARN) as an Express machine: it reached `SUCCEEDED`, the input propagating through all four steps, billed 500 ms / 64 MB; the resources were then torn down. A verified `STEPCHECK` workflow runs on the unmodified industrial platform.

6.10 Cross-format generalization: CNCF Serverless Workflow

To test generalization beyond AWS, we added a third frontend for the vendor-neutral CNCF Serverless Workflow DSL [7] and ran it on **66** example workflows from the specification’s repository. The frontend lowers the DSL’s tasks (`call/do/fork/switch/try/...`) to the *same* IR *without changing the IR or any analysis pass*, and parses all 66 with no failures. The structural, typestate, retry-safety, compensation, concurrency, and temporal analyses apply unchanged; only the ASL-specific JSONPath references the data-flow pass mines stay silent—correctly, since CNCF uses runtime expressions, not JSONPath, so `SC1101` and the binding checks (`SC1001–SC1003`) do not fire. A mutation study detects every injected defect with an injection site (retry 4/4, structural 9/9, concurrency 1/1, temporal 50/50). Crucially, CNCF can declare JSON Schemas, so where it does the typed contract check (`SC1010`) runs at *full precision with no inference*: a reordered schema-annotated order workflow is rejected with three `SC1010` errors from the declared schemas—the same defect inference only approximates on raw ASL, here exact. The asymmetry is informative: CNCF makes data contracts explicit but, like ASL, leaves idempotency and compensation implicit, so those checks still rely on annotation or inference in both ecosystems.

6.11 Threats to validity

Construct. The mutation study reports 100% recall *by design*: each mutation targets a defect within a check’s remit, so the result attests to the reliability of each check on its own defect class, not to coverage of all possible bugs—and we headline it only with that scope. We report it alongside the in-the-wild findings, which are not constructed. *Internal (gold labels).* The idempotency/persistence gold labels are human judgements; the 13 workflows added in this revision were labelled by two independent annotators under a written rubric with perfect agreement ($\kappa = 1.00$ for both properties), the original 10 were hand-labelled by an author, and we release all labels, the rubric, and the scoring harness. The contested cases (cancel, refund, keyed writes) are exactly where the heuristic errs against the human labels, and we report the resulting warning precision rather than hide it. *External.* The corpus is drawn from AWS sample repositories, which may over-represent didactic workflows; we mitigate this by taking *all* parseable workflows rather than a curated subset, yielding a realistic spread of sizes and features. *Soundness.* The data-flow and structural checks are conservative (they flag only definite faults), corroborated by zero `SC1101` false positives on the corpus; typestate and typed-contract precision depend on annotation quality, which is why `STEPCHECK` reports inferred findings as warnings. The concurrency and

temporal-budget analyses are *warnings*: the concurrency check is deliberately conservative (only last-writer-wins overwrites, never commuting or item-keyed writes), which is why it raises zero in-the-wild false positives while still detecting every injected race; a missing callback timeout (SC6001) may occasionally be acceptable for a short-lived task, so it too prompts review rather than asserting a bug. The budget check SC6002 did not fire in the wild (no corpus workflow declares a machine timeout tight enough to be overrun); we keep it as a defensive check and leave a labelled-precision study of the warning classes to future work.

7 Related Work

Serverless orchestration, durable execution, and transactions. The commercial orchestrators and their trade-offs have been analysed extensively [26,14], and many programming models make orchestration more expressive or data-aware [6]. Durable-execution engines (Temporal, Cadence [23,25]) and systems like Beldi [27] provide exactly-once execution and transactional/Saga semantics *at run time*, via logs and recovery protocols. All target execution—performance, scalability, convenience, fault tolerance. STEP CHECK is complementary: it targets *correctness*, statically, on the most widely deployed platform, adding no runtime machinery; a verified workflow can still run on a durable backend.

Typestate, sessions, and Saga compensation. Typestate originates with Strom and Yemini [21] and extends to objects and API protocols [10]; session and multiparty session types give a rich theory of protocol conformance [13]. STEP CHECK adapts the core idea—valid-state-only composition—to workflow tasks, trading full generality for a decidable, linear-time check inside an industrial toolchain. The Saga pattern [12] underpins cloud-native recovery [15], but compensation is usually coded and validated at run time [8]; STEP CHECK adds a *static* completeness check—every persistent operation must have a reachable compensator before shipping.

Data-flow analysis and ASL tooling. The validators available for ASL—`statelint` [3], `asl-validator`, and AWS’s `ValidateStateMachineDefinition`—check the J2119/ASL JSON schema, transition existence, and JSONPath *syntax*; none reasons about field provenance. Our data-flow pass casts ASL’s I/O-processing pipeline as an abstract interpretation [9] over a may-present lattice of document shapes, yielding a *sound* (no-false-positive) provenance check—to our knowledge the first *sound, shape-level* data-flow analysis for a deployed serverless workflow language, and the first of any kind for ASL beyond JSONPath syntax (it builds on, and is distinguished from, the business-process data-flow tradition discussed next). It is the workflow-internal analogue of the producer/consumer reasoning that consumer-driven contracts and static REST-client checks apply *across* services [18]; we make this analogy precise—the typed tier is a sound static consumer-driven-contract check between orchestrated tasks.

Data-flow verification of workflow languages. Detecting that a task reads data no predecessor produces is a long-studied *data-flow anomaly*. Sun et al. [22] formulate the data-flow perspective for business processes and detect *missing*, *redundant*, and *conflicting* data, and Trcka et al. [24] formalize the “missing data” anti-pattern (a datum read but never written) together with its parallel-branch read/write variant, discharged by CTL* model checking over workflow nets with data; WS-BPEL gained SSA/CSSA-based static data-flow analysis of uninitialized-variable reads [16], and data-flow-error checking has been specialized to BPMN 2.0 [20]. STEP CHECK targets the same *class* of defect but differs in technique and reach: prior work abstracts data to opaque named elements and checks a hand-built net by model checking (with the attendant risk of false positives and state-space explosion), whereas STEP CHECK tracks concrete *nested JSON document shapes* through ASL’s exact `InputPath` $\rightarrow$ `Parameters` $\rightarrow$ `ResultSelector` $\rightarrow$ `ResultPath` $\rightarrow$ `OutputPath` pipeline as a soundness-by-construction abstract interpretation [9], and it runs on—and recompiles to—the *deployed* artifact rather than a model of it. The document-shape lattice is in the spirit of shape inference over structured data [17]. On idempotency, FLUX [11] *verifies* idempotence of serverless function *code* by SMT reasoning; STEP CHECK instead *infers* idempotency at the orchestration layer from naming/resource conventions and uses it for retry-safety warnings without function-body access—complementary, not competing.

Concurrency and workflow verification. Interference and coordination for concurrent objects and distributed transactions are studied for actors and state-machine DSLs—interference-free network objects, and path-sensitive atomic commit that reasons about whether two messages’ effects commute [19]—but not for serverless-workflow `Parallel/Map` fan-out, which our concurrency pass targets. The closest workflow-level precedent is Trcka et al.’s parallel data-flow anti-pattern (a datum updated while a concurrent branch may read it) [24]; our check differs in targeting the serverless *resource* model—last-writer-wins `PutItem/PutObject` versus commuting `UpdateItem` and fire-and-forget publishes—natively on raw ASL. Workflow soundness is verified with Petri nets [1,5] and service compositions with model checking [4]: strong, general guarantees that require dedicated formalisms and are rarely wired into everyday cloud development. STEP CHECK deliberately targets a deployed industrial language with lightweight, linear checks and direct compilation. Against representative approaches—AWS Step Functions [2] and CNCF [7] (no static semantic checks), Temporal/Cadence [23,25] and Beldi [27] (runtime retry/Saga, not static), Petri-net/BPEL verification [5,4] (static but heavyweight, non-AWS), and session/typestate types [13,21] (typestate only)—STEP CHECK is, to our knowledge, the first to combine sound data-flow provenance, typestate conformance, retry safety, concurrency-interference and temporal-budget checking, and compensation completeness in a single *static* verifier that compiles directly to executable AWS Step Functions.

8 Conclusion

Serverless workflow orchestration has become a backbone of cloud-native systems, yet its workflows are untyped configuration whose correctness is checked only when they run—and the tooling developers have stops at JSONPath *syntax*. We presented STEP CHECK, a static verifier whose centrepiece is the first *sound*, shape-level data-flow analysis for a deployed serverless workflow language: a flow-sensitive abstract interpretation over a may-present lattice of document shapes that catches missing-field references with a proved no-false-positive guarantee and no annotations. Around it STEP CHECK adds seven further checks—structural, typed contracts, typestate, retry safety, concurrency interference, retry/timeout budget, and Saga compensation—two of them for classes no ASL tool had addressed, all compiling back to executable ASL. The key to analysing the workflows that already exist is a two-layer model: sound native checks that need nothing beyond ASL, paired with a lightweight annotation layer whose defaults are inferred from the naming and resource conventions that pervade real workflows.

Our evaluation on 193 real Step Functions workflows (plus 66 CNCF) supports the three research questions. STEP CHECK detects the targeted defect classes (100% recall over 616 injected defects) where a standard schema validator detects none of the retry, compensation, concurrency, or temporal defects, surfaces 33 unbounded-callback defects in unmodified workflows, and catches four genuine defects in mined real-world fix commits; the sound data-flow analysis fires zero false positives in the wild, as does the conservative concurrency check (RQ1). It recovers persistence at 87% and idempotency at 70% accuracy at zero authoring cost against a human-labelled gold set (two independent annotators, $\kappa = 1.00$; Section 6.11)—enough to make the compensation warnings mostly actionable (70% precision for SC4001, 100% for the effect-aware SC4010)—all surfaced as correctable warnings, and its typed tier recovers two-thirds of injected missing-field defects, the remainder being ASL’s intrinsic opacity (RQ2); and it verifies a workflow in $\approx 39 \mu s$ while adding zero runtime overhead, with verified workflows running unmodified on real AWS Step Functions (RQ3).

The broader message mirrors the history of type systems: just as static typing moved classes of error from run time to compile time, lightweight static verification—rooted in a sound data-flow core—can do the same for serverless orchestration, on the platforms developers already use. Future work includes further frontends (SAM, CDK), richer effect/ownership information to sharpen the compensation and concurrency analyses, a precise element-level shape for `Map/Parallel` item contexts, and a labelled-precision study of the concurrency and temporal warnings. We release STEP CHECK, the corpus, the gold labels and labelling rubric, and the evaluation harness to support replication and reuse, with an anonymized artifact link provided for review.

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